Hina Suzuki

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Research Interests

Computational black-hole astrophysics; strong-field gravity; black-hole imaging; general relativistic radiative transfer and magnetohydrodynamic simulations; high-performance computing; quantum computing.

Education

Heriot-Watt University Edinburgh, UK
Ph.D. in Electrical Engineering 2026 – Present

The University of Arizona Tucson, AZ, USA
B.S. in Astronomy, minor in Electrical and Computer Engineering; magna cum laude; GPA: 3.77/4.00 2022–2025

Research Experience

Center for Astrophysics | Harvard & Smithsonian Cambridge, MA
BHEX Summer Internship | Advisors: Filip Niewinski and Dr. Daniel Palumbo June 2026–July 2026
Black Hole Parameter Inference with Geometric Model Fitting

- Applied geometric model-fitting techniques to investigate black-hole parameter inference.
- Presented the project in the BHEX Summer Intern Final Presentation at the Center for Astrophysics | Harvard & Smithsonian.

Event Horizon Telescope Data & Science, Steward Observatory, The University of Arizona Tucson, AZ
Undergraduate Researcher | Advisors: Dr. Chi-Kwan Chan and Dr. Yosuke Mizuno January 2023–May 2025
High-Order Photon Ring Imaging of Kerr Naked Singularities

- Developed a pipeline to analyze photon-ring geometry around Kerr naked singularities using general relativistic radiative transfer calculations on high-performance computing resources.
- Generated high-resolution images from GRMHD-simulated data across viewing geometries and shadow morphologies.
- Found cascaded photon-ring structure, including rings inside the nominal shadow region, as a distinctive feature of the modeled naked singularities.
- Work resulted in a first-author publication, *High-Order Photon Rings around Kerr Naked Singularities*.

A Direct Calibration Algorithm for ADC Interleaving | Advisor: Dr. Chi-Kwan Chan

- Refined a calibration approach for mismatches in interleaved analog-to-digital converter systems using Fourier-transform methods.
- Performed numerical simulations to evaluate calibration accuracy with and without modeled system noise; the work was motivated by Event Horizon Telescope instrumentation and has applications in telecommunications and medical imaging.
- Contributed to *A Direct Calibration Algorithm for ADC Interleaving*, submitted to *The Astrophysical Journal*.

Steward Observatory Radio Astronomy Laboratory, The University of Arizona Tucson, AZ
Undergraduate Researcher | Advisor: Dr. Christopher Walker January 2024–December 2025
3 m Inflatable Aperture Telescope Mechanical Design and Fabrication

- Fabricated a 3 m reflector using a 1 mil membrane and its mounting structure for vacuum-tolerance testing.
- Performed metrology on the inflated reflector and characterized metrology behavior as a function of pressure.
- Designed and reviewed radio-frequency transmission circuits for the detector; work is associated with a co-authored paper in preparation.

Publications

Suzuki, H., ... & Chan, C.-K., et al. *High-Order Photon Rings around Kerr Naked Singularities*. Article link.

Chan, C.-K., **Suzuki, H.**, ... West, A. T., et al. *A Direct Calibration Algorithm for ADC Interleaving*. arXiv:2511.20763; manuscript submitted to *The Astrophysical Journal*.

Quantum Computing and Software Contributions

JIJ Inc. Tokyo, Japan
Quantum Software Engineering Intern August 2026–September 2026
Qamomile–Cirq Transpiler Contribution

- Implemented Cirq as a transpilation target/backend for Qamomile, an open-source quantum programming framework.
- Developed support for converting Qamomile quantum programs into Cirq circuits, expanding interoperability with the Cirq ecosystem.

Engineering and Research Employment

FreeFall Aerospace Tucson, AZ
Engineering Technician May 2025–May 2026

- Assisted development of a precision free-fall experimental system for gravitational testing through machining, assembly, CAD modification, and fabrication of mechanical components.
- Developed motor-control software to integrate mechanical and computational systems for experimental automation.

National Radio Astronomy Observatory, Electronics Department Socorro, NM
Electrical Engineering Intern | Supervisor: David Schafer May 2024–August 2024

- Rehabilitated motor-control electronics for a 3 m dish antenna at the Very Large Array and refined electrical wiring documentation.
- Developed a graphical user interface for online antenna control and helped establish an interface between a direction-finding antenna and an environmental monitoring system for radio-interference investigations.
- Upgraded the VLA visitor-center solar radio telescope: designed comparator and analog-to-digital circuits with operational amplifiers, assembled a PCB and LED signal display, and designed and coordinated fabrication of a 3D-CAD solar finder.

Selected Presentations

BHEX Summer Intern Final Presentation, Center for Astrophysics | Harvard & Smithsonian July 2026
Black Hole Parameter Inference with Geometric Model Fitting

Harvard ITC Luncheon, Institute for Theory and Computation, Harvard February 2026
High-Order Photon Rings around Naked Singularities

247th AAS Meeting, Phoenix, AZ January 2026
High-Order Photon Rings around Naked Singularities

Steward Symposium, Steward Observatory, The University of Arizona November 2024
A Direct Calibration Algorithm for ADC Interleaving

Awards and Honors

Department of Astronomy Outstanding Senior Award, The University of Arizona 2025
Undergraduate Research Travel Grant, The University of Arizona 2025
Galileo Circle Scholarship, The University of Arizona
(top 1.5% of College of Science undergraduate and graduate students) 2025
Angelos C. Langadas Scholarship in Astronomy, The University of Arizona 2025
William F. Lucas Astronomy Scholarship, The University of Arizona 2024
Wildcat Tuition Award, The University of Arizona 2022

Technical Skills

Programming: Python, C/C++, MATLAB, Bash.
Scientific computing: JAX, PyTorch, high-performance computing, numerical simulation, Git, LaTeX.
Quantum computing: Qiskit, Cirq, Qamomile.
Engineering: SolidWorks, KiCad, PCB assembly, CAD, spectrum analyzer, analog-to-digital converter systems, oscilloscope, multimeter.

Conferences and Workshops

Black Hole Explorer Science Meeting, virtual September 2025
AstroTech Summer School, Department of Astronomy, University of California, Berkeley June 2025
Black Hole Explorer Japan Workshop, virtual, National Astronomical Observatory of Japan June 2024

Languages

English (fluent); Japanese (native).